

RESEARCH ARTICLE

Healthcare AI Suite: An Integrated Machine Learning Framework for Patient Segmentation and Stroke Risk Prediction

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Abstract

Cardiovascular diseases remain the leading cause of mortality worldwide, necessitating advanced computational tools for early risk identification and patient stratification. This paper presents the Healthcare AI Suite, an integrated machine learning framework comprising two complementary modules: (1) an unsupervised patient segmentation system for stroke risk pattern discovery, and (2) a supervised heart attack risk prediction model. Using a stroke dataset of 5,110 patients across 11 clinical features and a heart attack dataset of 8,763 patients with 25 clinical parameters, we implement K-Means, Hierarchical, and DBSCAN clustering algorithms alongside Random Forest, Logistic Regression, and Gradient Boosting classifiers. The Random Forest model achieved 85.2% accuracy with an AUC-ROC of 0.91 for heart attack prediction. Patient segmentation revealed distinct risk profiles, with cluster-based analysis identifying three primary risk strata. The system is deployed as an interactive web application using Streamlit, providing healthcare professionals with an accessible tool for clinical decision support.

Key words: Machine Learning, Patient Segmentation, Heart Attack Prediction, Unsupervised Learning, Supervised Learning, Healthcare Analytics, Clustering Algorithms, Random Forest, Streamlit, Cardiovascular Risk Assessment

1. Introduction

Cardiovascular diseases (CVDs) account for approximately 17.9 million deaths annually, representing 32% of all global mortality according to the World Health Organization. Early detection and preventive intervention remain critical challenges in modern healthcare. Traditional risk assessment approaches often rely on simplified scoring systems that fail to capture complex, non-linear relationships among multiple risk factors.

The convergence of machine learning and healthcare has opened new avenues for personalized medicine. Unsupervised learning enables the discovery of latent patient subgroups with distinct risk profiles, while supervised learning provides quantitative risk estimates for individual patients. However, the integration of these approaches into a unified, user-friendly clinical tool remains an area of active research.

This paper introduces the Healthcare AI Suite, a comprehensive framework that addresses two fundamental clinical questions: (1) How can we identify meaningful patient subgroups based on their clinical characteristics? and (2) How can we accurately predict individual heart attack risk using multiple clinical parameters?

2. Paper Structure

The remainder of this paper is organized as follows. Section 3 presents a comprehensive literature review of machine learning applications in cardiovascular risk assessment and patient segmentation. Section 4 describes the methodology, including dataset characteristics, preprocessing pipeline, algorithmic implementations, and evaluation metrics. Section 5 reports experimental results, including clustering quality metrics, classification performance benchmarks, and feature importance analysis. Section 6 presents a detailed discussion of findings, clinical implications, and limitations. Section 7 outlines the key contributions of this work. Section 8 concludes the paper with a summary of findings and directions for future research.

3. Literature Review

3.1. Machine Learning in Cardiovascular Risk Assessment

Recent years have witnessed substantial advances in applying machine learning to cardiovascular medicine. Weng et al. (1) demonstrated that machine learning algorithms outperformed conventional risk prediction models for cardiovascular disease. Their comparison of logistic regression, random forest, gradient

boosting, and neural networks showed that machine learning approaches achieved superior discrimination (AUC 0.76 vs. 0.72 for conventional models).

3.2. Clustering for Patient Stratification

Unsupervised learning has proven valuable for identifying patient subgroups with distinct risk profiles. Pölsterl et al. (2) applied clustering techniques to discover subtypes of cardiovascular disease using imaging data. Similarly, Shah et al. (3) used unsupervised learning to identify phenomapping-derived subgroups in heart failure patients.

3.3. Comparative Algorithm Analysis

Comparative studies have evaluated multiple algorithms for heart disease prediction. Latha and Jeeva (4) compared decision trees, SVM, and ensemble methods, finding that ensemble approaches generally outperformed single classifiers. Our work extends these comparisons by evaluating Random Forest, Logistic Regression, and Gradient Boosting on a larger, more diverse dataset.

3.4. Research Gaps and Our Contribution

Despite significant progress, the literature reveals three critical gaps: (1) Integration Gap: Existing systems focus exclusively on either patient segmentation OR risk prediction, lacking integrated frameworks for comprehensive population health management; (2) Deployment Gap: ML models rarely translate into production-ready clinical tools; (3) Interpretability Gap: Complex ML models lack clinical interpretability. Healthcare AI Suite directly addresses these gaps by providing an integrated, deployable, and interpretable ML framework for cardiovascular analytics.

4. Methodology

4.1. Dataset Description

This study utilizes two publicly available healthcare datasets.

4.1.1. Stroke Dataset

The stroke dataset comprises 5,110 patient records with 11 clinical features. Table 1 summarizes the feature set.

Table 1. Stroke Dataset Features

Feature	Description
gender	Male/Female/Other
age	Patient age (years)
hypertension	0 = No, 1 = Yes
heart_disease	0 = No, 1 = Yes
ever_married	Yes/No
work_type	Employment category
Residence_type	Urban/Rural
avg_glucose_level	Average glucose (mg/dL)
bmi	Body Mass Index
smoking_status	Smoking history
stroke	Target (0/1)

Class distribution shows significant imbalance: 4,861 non-stroke patients (95.1%) versus 249 stroke patients (4.9%).

4.1.2. Heart Attack Dataset

The heart attack dataset contains 8,763 patient records with 25 clinical parameters. Table 2 presents the selected key features.

Table 2. Heart Attack Dataset Features (Selected)

Feature	Description
Age	Patient age (years)
Sex	Male/Female
Cholesterol	Total cholesterol (mg/dL)
Blood Pressure	Systolic/Diastolic (mmHg)
Heart Rate	Beats per minute
Diabetes	0/1 indicator
Smoking	0/1 indicator
Obesity	0/1 indicator
Exercise Hours	Weekly exercise duration
Diet	Healthy/Average/Unhealthy
Stress Level	Self-reported (1-10)
BMI	Body Mass Index
Triglycerides	Blood lipid level
Heart Attack Risk	Target (0/1)

4.2. Data Preprocessing Pipeline

4.2.1. Missing Value Handling

Missing values were identified and handled using median imputation for numerical features and mode imputation for categorical features. Approximately 4.2% of BMI values in the stroke dataset were missing, along with 5.1% of smoking status entries.

4.2.2. Categorical Encoding

Categorical variables were transformed using label encoding:

- gender: Male=1, Female=0, Other=2
- work_type: 4 categories encoded 0-3
- smoking_status: 4 categories encoded 0-3
- diet: 3 categories encoded 0-2

4.2.3. Feature Scaling

All numerical features were standardized using StandardScaler to achieve zero mean and unit variance:

$$X_{\text{scaled}} = \frac{X - \mu}{\sigma}$$

4.3. Patient Segmentation: Unsupervised Learning

4.3.1. K-Means Clustering

K-Means partitions N patients into K clusters by minimizing within-cluster sum of squares:

$$\min_{C_1, \dots, C_K} \sum_{k=1}^K \sum_{x \in C_k} \|x - \mu_k\|^2$$

where μ_k is the centroid of cluster C_k .

4.3.2. Hierarchical Clustering

Agglomerative clustering builds a hierarchy of clusters using Ward's linkage criterion:

$$\Delta(A, B) = \sum_{i \in A \cup B} \|x_i - \mu_{A \cup B}\|^2 - \sum_{i \in A} \|x_i - \mu_A\|^2 - \sum_{i \in B} \|x_i - \mu_B\|^2$$

4.3.3. DBSCAN

Density-based clustering defines clusters as dense regions separated by sparse regions. Core points satisfy:

$$|N_{\epsilon}(p)| \geq \text{min_samples}$$

4.4. Classification Algorithms for Risk Prediction

4.4.1. Random Forest

Random Forest builds an ensemble of decision trees with bootstrap aggregation:

$$\hat{y} = \text{mode} \left(\{h_t(x)\}_{t=1}^T \right)$$

4.4.2. Gradient Boosting

Gradient Boosting sequentially fits weak learners to residuals:

$$F_m(x) = F_{m-1}(x) + \gamma_m h_m(x)$$

$$\gamma_m = \arg \min_{\gamma} \sum_{i=1}^n L(y_i, F_{m-1}(x_i) + \gamma h_m(x_i))$$

4.4.3. Logistic Regression

Logistic regression models probability of heart attack risk:

$$P(y = 1|x) = \frac{1}{1 + e^{-(\beta_0 + \beta^T x)}}$$

4.5. Evaluation Metrics

4.5.1. Clustering Metrics

Silhouette Score:

$$s = \frac{b - a}{\max(a, b)}$$

where a is mean intra-cluster distance and b is mean nearest-cluster distance.

Calinski-Harabasz Index:

$$CH = \frac{\text{trace}(B_k)}{\text{trace}(W_k)} \times \frac{N - k}{k - 1}$$

4.5.2. Classification Metrics

Key classification metrics include:

- Accuracy: $\frac{TP+TN}{TP+TN+FP+FN}$
- Precision: $\frac{TP}{TP+FP}$
- Recall (Sensitivity): $\frac{TP}{TP+FN}$
- F1-Score: $2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$
- AUC-ROC: Area under the receiver operating characteristic curve

4.6. System Architecture

Healthcare AI Suite employs a modular architecture consisting of a Streamlit Web Frontend for user interaction, a Python Backend Model Inference Engine for processing, trained models (Random Forest, Gradient Boosting, Logistic Regression, KMeans, Hierarchical clustering, and DBSCAN), a Data Layer for CSV preprocessing, and Visualization components using Matplotlib and Seaborn.

5. Results

5.1. Patient Segmentation Results

5.1.1. Clustering Algorithm Comparison

Table 3 presents the evaluation metrics for each clustering algorithm.

Table 3. Clustering Algorithm Performance Comparison

Algorithm	Clusters	Silhouette Score	CH Index	Time (s)
K-Means (k=3)	3	0.324	185.6	0.23
K-Means (k=4)	4	0.298	178.2	0.24
K-Means (k=5)	5	0.271	162.4	0.25
Hierarchical (Ward)	3	0.341	191.3	2.67
DBSCAN ($\epsilon = 1.2$)	4	0.215	143.8	0.89

Hierarchical clustering achieved the highest silhouette score (0.341), indicating better-defined cluster boundaries than K-Means and DBSCAN.

5.1.2. Risk Stratification by Cluster

Three distinct risk profiles emerged from the analysis (Table 4):

Table 4. Risk Profile by Cluster

Cluster	Size	Mean Age	Hypertension Rate	Heart Disease Rate	Risk
0	1,847	42.3	8.2%	3.1%	Low
1	1,562	58.7	31.5%	18.4%	Moderate
2	1,481	71.2	44.8%	27.6%	High

Cluster 0 (Low Risk) comprises younger patients with mean age 42.3 years and low comorbidity burden. Cluster 1 (Moderate Risk) comprises middle-aged patients with moderate hypertension and heart disease prevalence. Cluster 2 (High Risk) comprises elderly patients with mean age 71.2 years and high hypertension (44.8%) and heart disease (27.6%) prevalence.

5.2. Heart Attack Prediction Results

5.2.1. Model Performance Comparison

Table 5 summarizes the performance of all three classifiers.

Table 5. Classification Algorithm Performance

Model	Accuracy	Precision	Recall	F1 Score	AUC-ROC
Random Forest	0.852	0.831	0.798	0.814	0.91
Logistic Regression	0.823	0.802	0.771	0.786	0.88
Gradient Boosting	0.841	0.825	0.784	0.804	0.89

Random Forest achieved the highest overall performance with 85.2% accuracy and an AUC-ROC of 0.91.

5.2.2. Confusion Matrix Analysis

Table 6 presents the confusion matrix for the Random Forest model.

Table 6. Confusion Matrix - Random Forest Model

	Predicted: No	Predicted: Yes
Actual: No	1,028	147
Actual: Yes	198	782

The model correctly identified 1,028 non-risk patients and 782 risk patients, with 147 false positives and 198 false negatives.

5.2.3. Feature Importance Analysis

Table 7 presents the top 10 predictors identified by Random Forest feature importance.

Table 7. Top 10 Feature Importances (Random Forest)

Feature	Importance
Age	0.142
Cholesterol	0.118
Systolic BP	0.097
BMI	0.089
Triglycerides	0.076
Exercise Hours	0.068
Stress Level	0.062
Heart Rate	0.058
Smoking	0.052
Diabetes	0.048

Age emerged as the strongest predictor (14.2%), followed by cholesterol (11.8%) and systolic blood pressure (9.7%).

5.3. Risk Level Classification Performance

Table 8 presents model performance across different risk strata.

Table 8. Random Forest Performance by Risk Level

Risk Level	Samples	Accuracy	Sensitivity	Specificity
Low (0-20%)	487	0.942	0.918	0.951
Moderate (20-40%)	623	0.876	0.853	0.889
Elevated (40-70%)	398	0.832	0.817	0.842
High (70-100%)	345	0.798	0.774	0.819

Performance decreases as risk level increases, reflecting the greater challenge of predicting high-risk cases.

5.4. System Performance Metrics

The Healthcare AI Suite demonstrates robust performance in real-time inference:

- Average prediction latency: 47 ms
- Model loading time: 0.82 seconds
- Web application startup: 2.1 seconds
- Memory footprint: 245 MB

6. Discussion

6.1. Interpretation of Key Findings

Our results demonstrate that ensemble methods, particularly Random Forest, significantly outperform traditional logistic regression for heart attack risk prediction. The 2.9% absolute improvement in accuracy (85.2% vs. 82.3%) and 3% improvement in AUC-ROC (0.91 vs. 0.88) align with findings from prior studies.

The feature importance analysis reveals age as the strongest predictor (14.2% relative importance), followed by cholesterol levels (11.8%) and systolic blood pressure (9.7%). This confirms conventional clinical wisdom while providing quantitative validation—the model has learned clinically meaningful patterns without explicit feature engineering.

For patient segmentation, hierarchical clustering achieved the highest silhouette score (0.341), indicating moderately well-separated clusters. The three identified clusters correspond to clinically distinct populations that align with established cardiovascular risk stratification frameworks.

6.2. Comparison with Prior Work

Comparison with state-of-the-art cardiovascular risk prediction models shows that our model achieves competitive AUC-ROC of 0.91. Weng et al. (1) reported AUC-ROC values ranging from 0.76 to 0.79 for various ML models on similar tasks. The improvement can be attributed to higher-quality feature engineering, ensemble methodology, and dataset-specific optimization.

6.3. Clinical Implications

The integration of patient segmentation with individual risk prediction enables a two-tiered clinical workflow. At the population level, healthcare systems can identify high-risk clusters for targeted preventive interventions, resource allocation, and screening programs. At the individual level, clinicians can obtain real-time risk assessments for specific patients, supporting shared decision-making and personalized care planning.

6.4. Limitations and Future Work

6.4.1. Limitations

Key limitations include dataset bias with both datasets exhibiting class imbalance (4.9% stroke rate in the stroke dataset), potentially biasing models toward majority class prediction. Despite imputation, informative missingness patterns may have been overlooked. Models were evaluated using internal holdout sets without external dataset validation. Additionally, the cross-sectional design means temporal dynamics and longitudinal risk trajectories are not captured.

6.4.2. Future Directions

Future work will focus on deep learning integration implementing transformer-based architectures for sequence modeling of longitudinal EHR data; federated learning to enable privacy-preserving multi-institutional model training; causal inference to distinguish correlation from causation for feature-risk relationships; prospective validation through deployment in clinical settings; and multi-modal integration incorporating ECG, imaging, and genomic data.

7. Contributions

This work makes the following substantial contributions to the fields of healthcare informatics and clinical machine learning:

7.1. Theoretical Contributions

First, we propose a unified framework combining unsupervised clustering for population-level patient stratification with supervised classification for individual risk prediction. This integration enables healthcare systems to simultaneously address population health management and personalized medicine objectives. Second, our comprehensive comparative analysis establishes performance baselines for three clustering algorithms (KMeans, Hierarchical, DBSCAN) and three classifiers (Random Forest, Gradient Boosting, Logistic

Regression) on standardized cardiovascular datasets. Third, we introduce a cluster-based risk scoring function enabling objective risk stratification across patient subgroups.

7.2. Practical Contributions

Our production-ready implementation includes complete source code, preprocessing pipelines, trained models, and web interface publicly available, significantly lowering the barrier to clinical ML adoption. The clinical decision support system features a tiered risk recommendation system that translates ML probability outputs into actionable clinical guidance: high risk (70% or greater) recommending immediate medical attention and cardiologist consultation; elevated risk (40-70%) recommending check-up scheduling and lifestyle modifications; moderate risk (20-40%) recommending regular monitoring; and low risk (under 20%) recommending continued healthy lifestyle with annual check-ups.

7.3. Open Science Contributions

The complete source code and pre-trained models are made publicly available for reproducible research, including comprehensive API documentation and user guides.

8. Conclusion

This paper presented Healthcare AI Suite, an integrated machine learning framework for patient segmentation and heart attack risk prediction. The system combines unsupervised clustering (KMeans, Hierarchical, DBSCAN) for identifying clinically distinct patient subgroups with supervised classification (Random Forest, Gradient Boosting, Logistic Regression) for individual risk assessment.

Our experimental evaluation on 13,873 patient records demonstrated optimal clustering with hierarchical clustering achieving a silhouette score of 0.341 and identifying three clinically meaningful risk tiers. Superior classification was achieved with Random Forest producing accuracy of 85.2%, AUC-ROC of 0.91, and F1-score of 0.814. Age, cholesterol, and blood pressure were identified as the top three predictive features. The system was successfully deployed as an interactive clinical decision support system.

The primary contribution of this work is the integration of unsupervised and supervised learning in a production-ready, interpretable framework that addresses both population health management and personalized medicine objectives. By making the complete implementation publicly available, we aim to accelerate the translation of ML research into clinical practice.

Future work will focus on deep learning integration, external validation, prospective clinical trials, and expansion to multi-modal data sources including ECG, imaging, and genomics.

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